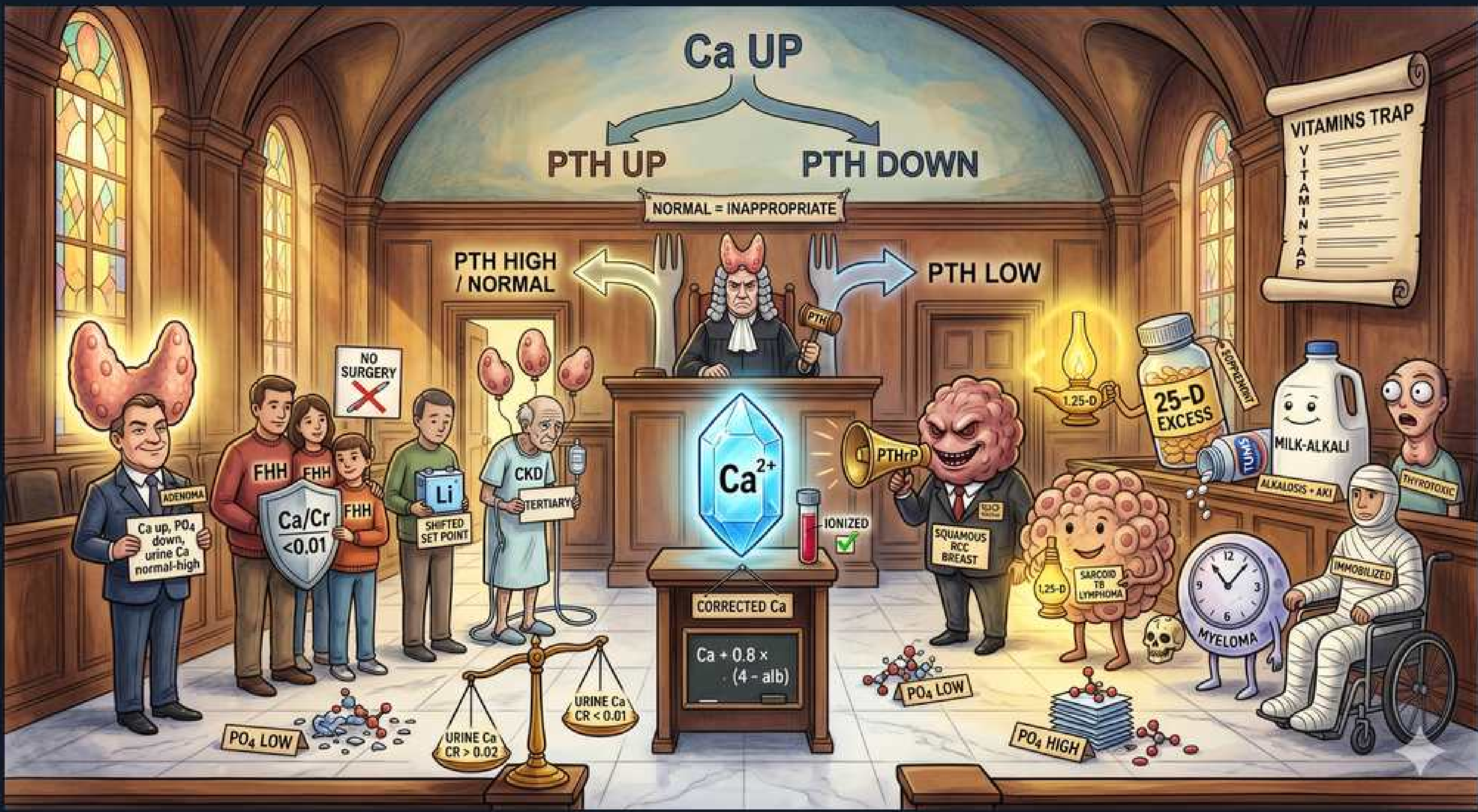
Calcium/PTH/VitD — The Calcium Courthouse



1. Ca UP banner

WHAT YOU SEE

Top banner CA UP feeding a judge who splits the case into PTH UP vs PTH DOWN — the courthouse 'first ruling' is: in any hypercalcemia, measure PTH first because it cleaves the entire differential in two.

WHAT IT ENCODES

The single most important first move in any hypercalcemia workup is to measure intact PTH, which dichotomizes every cause into PTH-mediated (high/inappropriately normal PTH) versus non-PTH-mediated (suppressed PTH). Confirm true hypercalcemia first by correcting for albumin or measuring ionized Ca. The two outpatient heavyweights are primary hyperparathyroidism (PTH up) and malignancy (PTH suppressed), which together cause >90% of cases.

BOARD TRAP

WRONG to jump to a malignancy workup (CT, mammogram, SPEP) before checking PTH. PTH is cheap and instantly splits the differential — a high/normal PTH makes malignancy very unlikely and points to parathyroid disease. Skipping PTH is the classic inefficient-workup trap.

---

2. PTH high/normal branch

WHAT YOU SEE

Left arrow off the judge reading PTH HIGH/NORMAL with the caption 'NORMAL = INAPPROPRIATE' — the signpost insists a non-suppressed PTH during high calcium is itself the crime (PTH-dependent causes).

WHAT IT ENCODES

A PTH that is frankly high OR even mid-normal in the face of hypercalcemia is INAPPROPRIATE — normal feedback should suppress PTH to near-undetectable when calcium is high. This 'inappropriately normal' PTH defines the PTH-dependent group: primary hyperparathyroidism (most common), familial hypocalciuric hypercalcemia (FHH), tertiary HPT, and lithium therapy.

BOARD TRAP

WRONG to call a PTH within the lab's normal reference range 'normal' and stop. With high calcium, a normal PTH is pathologically inappropriate. The discriminator is to interpret PTH RELATIVE to the calcium, not against the reference interval alone.

---

3. PTH low branch

WHAT YOU SEE

Right arrow off the judge reading PTH LOW — the signpost sends you away from the parathyroids toward malignancy and vitamin D causes, where PTH is correctly suppressed.

WHAT IT ENCODES

A suppressed (low) PTH with hypercalcemia means the parathyroids are correctly responding to a non-PTH source of calcium. Causes: malignancy (PTHrP, osteolytic mets, lymphoma-derived 1,25-D), vitamin D toxicity, granulomatous disease (sarcoid, TB), milk-alkali, thyrotoxicosis, immobilization, vitamin A excess, and Addisonian crisis. Next steps are PTHrP, 25-OH-D, and 1,25-D levels.

BOARD TRAP

WRONG to keep chasing parathyroid imaging (sestamibi/ultrasound) when PTH is suppressed — the parathyroids are innocent. Suppressed PTH redirects you entirely to malignancy and vitamin-D pathways. Sending a PTH-low patient to the surgeon is a category error.

---

4. Primary HPT figure

WHAT YOU SEE

Left witness figure holding enlarged parathyroid glands with a card 'Ca UP, PO4 down, urine Ca normal/high' — the over-grown glands picture the autonomous adenoma of primary HPT, the top outpatient cause.

WHAT IT ENCODES

Primary hyperparathyroidism is the most common cause of OUTPATIENT (asymptomatic, incidentally found) hypercalcemia, usually a single adenoma (~85%). Labs: high Ca, LOW phosphate (PTH causes renal PO4 wasting), high/inappropriately-normal PTH, high-normal/high urine calcium, and high 1,25-D. Mostly detected on routine chemistry in postmenopausal women.

BOARD TRAP

WRONG to confuse it with malignant hypercalcemia, the most common INPATIENT cause. Malignancy gives suppressed PTH, faster/higher calcium rise, and a sick patient; primary HPT gives non-suppressed PTH and a well, chronically mildly hypercalcemic patient. PTH level and clinical tempo discriminate.

---

5. FHH / Ca-Cr <0.01

WHAT YOU SEE

A whole family group holding an FHH sign with 'Ca/Cr <0.01' and a red-X 'NO SURGERY' — the family stands for the autosomal-dominant CaSR defect, and the low urine-calcium ratio is the clue that spares them the OR.

WHAT IT ENCODES

Familial hypocalciuric hypercalcemia is an autosomal dominant inactivating CaSR mutation that biochemically mimics mild primary HPT (high Ca, high/normal PTH) but is benign and lifelong. The separator is LOW urine calcium: 24-hour calcium-to-creatinine clearance ratio <0.01. No surgery, no treatment needed.

BOARD TRAP

WRONG to send an FHH patient to parathyroidectomy — surgery fails because the CaSR defect is body-wide, not in the gland, and calcium stays high. Always compute the urine Ca/Cr clearance ratio before referring: <0.01 = FHH (don't operate), >0.02 = primary HPT. This is THE high-yield discriminator.

---

6. Lithium / CKD figures

WHAT YOU SEE

Two figures under the PTH-high branch tagged Li (shifted set-point) and CKD (tertiary) — the lithium bottle shifts the gland's calcium thermostat, and the CKD figure shows glands gone autonomous after years of stimulation.

WHAT IT ENCODES

Lithium shifts the parathyroid calcium set-point upward (reduces CaSR sensitivity), causing hypercalcemia with high/normal PTH and low urine Ca — a drug-induced FHH-like picture. Tertiary hyperparathyroidism arises in long-standing CKD/secondary HPT when chronically stimulated glands become AUTONOMOUS, now over-secreting PTH and driving calcium HIGH despite the original hypocalcemic stimulus.

BOARD TRAP

WRONG to rush a lithium patient to parathyroid surgery before a drug trial off lithium (when psychiatrically safe) — lithium hypercalcemia can be reversible. And WRONG to confuse SECONDARY HPT (low/normal Ca, high PTH from CKD) with TERTIARY (HIGH Ca, very high autonomous PTH). The calcium level distinguishes secondary from tertiary.

---

7. Corrected Ca / ionized podium

WHAT YOU SEE

The central glowing blue Ca2+ crystal on the witness stand with an IONIZED vial and the chalkboard formula 'Ca + 0.8 x (4 - albumin)' — the radiant crystal is the true active (ionized) calcium you must correct to before judging.

WHAT IT ENCODES

About 40% of serum calcium is albumin-bound, so total calcium tracks albumin. Corrected Ca = measured Ca + 0.8 × (4.0 − albumin g/dL). The physiologically active fraction is IONIZED calcium, which is what the CaSR senses; when in doubt (low albumin, critical illness, acid-base derangement) measure ionized calcium directly.

BOARD TRAP

WRONG to diagnose hypocalcemia in a hypoalbuminemic patient (cirrhosis, nephrotic, malnutrition) from a low TOTAL calcium — only the protein-bound fraction fell, ionized calcium is normal and the patient is asymptomatic. Correct for albumin or measure ionized Ca before treating. Conversely, acidosis raises ionized Ca for a given total.

---

8. Vitamins trap scroll

WHAT YOU SEE

Top-right hanging scroll titled VITAMINS TRAP — the mnemonic parchment cataloguing the PTH-suppressed (non-parathyroid) causes of hypercalcemia.

WHAT IT ENCODES

The 'VITAMINS TRAP' mnemonic catalogs non-PTH (PTH-suppressed) hypercalcemia: Vitamin D/A excess, Immobilization, Thyrotoxicosis, Addison disease, Milk-alkali, Inflammatory/granulomatous (sarcoid, TB), Neoplasm, Sarcoidosis, Thiazides, Rhabdomyolysis (recovery), Aids/Adenoma overlap, Paget, Parenteral nutrition. Granulomas and lymphoma raise calcium via macrophage 1-alpha-hydroxylase producing 1,25-D.

BOARD TRAP

WRONG to forget that THIAZIDES (decrease urinary Ca excretion) and excess vitamin A can independently raise calcium. In a hypercalcemic patient, review the med list (thiazide, lithium, vitamin D/A, calcium-antacids) before an expensive workup — drugs are reversible causes hiding in plain sight.

---

9. PTHrP / malignancy monster

WHAT YOU SEE

A green sunglasses-wearing MONSTER under the PTH-low branch holding a PTHrP megaphone — the impostor monster mimics PTH's effect on bone/kidney but is a separate molecule, so the standard PTH assay reads LOW.

WHAT IT ENCODES

Malignancy is the most common cause of inpatient/severe hypercalcemia, via three mechanisms: (1) humoral — PTHrP from squamous cell (lung, head/neck), renal, breast, bladder cancers; (2) osteolytic metastases and myeloma releasing local cytokines/RANKL; (3) lymphoma producing 1,25-D. All show SUPPRESSED endogenous PTH. Acute management: aggressive IV normal saline, then a bisphosphonate (zoledronic acid) ± calcitonin for rapid bridging; denosumab if bisphosphonate-refractory or renal failure.

BOARD TRAP

WRONG to expect a standard PTH assay to detect PTHrP — they are different molecules; intact PTH is LOW while PTHrP must be ordered as a separate test. Concluding 'PTH normal so parathyroids fine, no cause found' misses humoral malignancy. Order PTHrP explicitly when PTH is suppressed.

---

10. 25-D excess / milk-alkali

WHAT YOU SEE

Right-side bottles labelled 25-D EXCESS and MILK-ALKALI (with ALKALOSIS + AKI tags) — the supplement and antacid bottles are the two intake-overload causes of PTH-suppressed hypercalcemia.

WHAT IT ENCODES

Vitamin D intoxication raises 25-OH-D (the storage form measured to confirm it) and increases gut absorption of BOTH calcium and phosphate, so it gives high Ca AND high PO4 with suppressed PTH. Milk-alkali syndrome (large calcium + absorbable alkali intake, e.g., calcium carbonate for dyspepsia/osteoporosis) classically gives the triad of hypercalcemia, metabolic ALKALOSIS, and acute kidney injury.

BOARD TRAP

WRONG to lump granulomatous hypercalcemia with vitamin D INTOXICATION — toxicity raises 25-OH-D, whereas sarcoid/lymphoma raise 1,25-D (calcitriol) via macrophage 1-alpha-hydroxylase with normal/low 25-D. Check WHICH vitamin D metabolite is elevated; granulomatous disease also responds to glucocorticoids.

---

11. Myeloma / immobilized patient

WHAT YOU SEE

A wheelchair-bound, bandaged IMMOBILIZED patient beside a MYELOMA skull with punched-out lesions — the immobile, lytic-boned figure shows resorption-driven hypercalcemia with suppressed PTH.

WHAT IT ENCODES

Multiple myeloma causes hypercalcemia through plasma-cell cytokine/RANKL-driven osteoclastic resorption (osteolytic 'punched-out' lesions), with suppressed PTH. Immobilization causes hypercalcemia (especially in high-turnover states like Paget or young patients) by uncoupling resorption from formation. Workup clues for myeloma: anemia, renal failure, high total protein/gap, SPEP/UPEP monoclonal spike.

BOARD TRAP

WRONG to read myeloma's lytic lesions as metastatic bone disease and order a bone scan — technetium bone scans are typically NEGATIVE in myeloma (no osteoblastic reaction). Use skeletal survey/low-dose whole-body CT or MRI plus SPEP/UPEP. Negative bone scan with lytic lesions points to myeloma.

---

12. Urine Ca / PO4 evidence

WHAT YOU SEE

Bottom balance-scale evidence: urine Ca/Cr ratio with PO4 LOW on the left pan vs PO4 HIGH on the right — the weighed phosphate and urine-calcium tip the verdict between the competing causes.

WHAT IT ENCODES

Ancillary biochemistry refines the PTH-high branch: primary HPT shows LOW serum PO4 (PTH wastes phosphate) with urine Ca/Cr clearance ratio >0.02; FHH shows LOW urine calcium (ratio <0.01). On the PTH-low branch, vitamin D toxicity raises both Ca and PO4. So the phosphate direction and the urine calcium together fingerprint the cause.

BOARD TRAP

WRONG to use a single spot urine calcium without the clearance ratio when separating FHH from primary HPT — the formula (UCa × SCr) / (SCa × UCr) corrects for renal function and diet. A naked spot urine Ca can mislead; the calculated Ca/Cr CLEARANCE ratio (<0.01 FHH, >0.02 primary HPT) is the validated cutoff.
